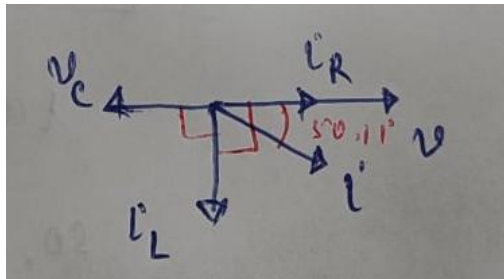


Tutorial_5_Answer Key

1. $i(t) = 0.7 \cos(5000t + 39.09^\circ) A$
2. $V_a(t) = 87.45 \cos(1000t - 47.73^\circ) V$
3. $i_1(t) = 1.05 \cos(100t + 71.56^\circ) A$
4. $v(t) = 4320 \cos(1000t + 135^\circ) V$
5. (a) $Z(j\omega) = (0.82 + j0.982) \Omega$
 (b) $i(t) = 5.51 \cos(5t - 50.11^\circ) A$
 $i_R(t) = 3.54 \cos 5t A$
 $i_L(t) = 4.23 \cos(5t - 90^\circ) A$



6. $S = (400 + j400) KVA$ (In Rectangular form)
 $565.69 \angle 45^\circ KVA$ (In Polar form)
7. (a) $I = 90.91 A$
 $Q = 15.18 kVAR$
 (b) $R = 1.21 \Omega$ and $C = 1.44 mF$
8. $R = 5 \Omega$ and $L = 31.8 mH$
9. $S_{total} = 27.93 kVA$ and $power\ factor = 0.967 (lag)$
10. $Q_C = 9.6 kVAR$